

Bazaar



We test the latest and the best hardware and software products available in the market

Pentium 4 3.06 GHz

Intel's newest, baddest and fastest desktop processor

Intel's Pentium 4 3.06 GHz processor finally makes HyperThreading a reality for the desktop PC. With this technology, your processor will appear like two separate CPUs to your OS. The current version of HyperThreading technology allows a maximum of two threads to be dispatched to the CPU. Everything else has still stayed the same. So really how good is the processor?



rather than a synthetic one and is designed to show the improvements that instruction sets used in various processors (MMX/Enhanced, 3DNow! /Enhanced, and SSE (2)) bring to such an algorithm. We also ran the Memory benchmark, which measures sustained memory bandwidth. Here the P4 3.06 GHz easily outperformed the 2.8 GHz-based processor even with HyperThreading turned off.

In the CPU multimedia subtest too, the HyperThreading-enabled CPU posted a higher score in the FPU SSE region though the integer scores don't offer much to talk about.

To test the gaming performance of the processor, we ran a time demo on the retail version of *Quake III Arena* patched with version 1.30. The test was run in three different settings: normal, high and max (1024x786x32x32).

In the normal mode there was no difference in the scores with HyperThreading switched on and off, but the difference was noticeable in the high and max quality modes.

Our second gaming test involved running *Comanche 4* in three different settings, normal (640x480x16), high (800x600x16) and maximum (1024x786x32). Here too there

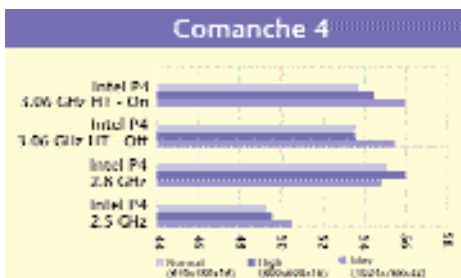
wasn't much difference in the frame rates with HyperThreading turned on and off.

Another area where a high amount of CPU power is required is video encoding. To measure the performance of a processor during video encoding, we encoded a standard MPEG-2 format video sequence using VirtualDub v1.4. The DivX codec used was v4.02 and the setting for audio and video processing in VirtualDub was turned to full processing mode. The video compression bit rate was set at 900 Kbps. Here the less time it

ple of benchmarks simultaneously. We ran the Povray benchmark, which is severely CPU intensive, while simultaneously running *Comanche 4* in the normal mode. We ran this test first with HyperThreading turned off and then with HyperThreading turned on just to see if it makes a difference—we achieved an average 26.57 fps with the HyperThreading disabled and a boost of approximately 11 fps with it enabled.

All in all, Intel's latest desktop processor truly impresses with its performance and the

HyperThreading feature makes it achieve higher performance scores under stress. HyperThreading in its future iterations could very well overcome its little drawbacks



takes to encode the file, the better it is. We recorded our lowest ever score—the CPU took a mere 51 seconds to encode the video file. This makes the P4 3.06 a great option for those who work with video encoding applications to a large extent.

Just to put the CPU under a bit more stress, we ran a cou-

ple that still exist—for instance, it could reduce performance if the OS dispatches multiple threads that are aimed at similar resources in the CPU. Until then this is as fast as you could get.

SPECIFICATIONS

0.13 micron fabrication, 478 pins, 533 MHz FSB bus, HyperThreading

Our first tests included synthetic benchmarks. Here we ran SiSoft Sandra Professional 2003 and noted the scores in different areas. In the CPU benchmark that indicates the theoretical performance potential of the CPU, the Pentium 4 3.06 GHz scored 9059 CPU Drystone marks and 5567 Whetstone marks.

The CPU multimedia subtest is a real-life benchmark

Price: Rs 41,500
Contact: Nebula Technologies Pvt Ltd
Phone: 022-22670567
Fax: 022-26331074
E-mail: pratik@nebulatech.com
Web site: www.intel.com

Pentium 4 3.06 GHz		B+
Performance	▶▶▶	